For all the following functions, we want to find out whether $\lim_{(x,y)\to(0,0)} f(x,y)$ exists, and find the limit if it does exist.

These are all trickier than you'd get in a test, but are good practice nonetheless, and they make it clear limits are not to be taken lightly!

Try these first on your own! These are all from Exercise 0.14 in Domain-Limits-continuity.

Exercises (k), (m), (g), and (n), (labelled as such in Ex.0,14) are relatively doable and offer great squeeze theorem practice

$$f(x,y) = \frac{xy^2 - x^2}{x^2 - 2xy + y^2}$$ substitution gives $\frac{0}{0}$

along x-axis?

$$f(x,0) = \frac{x \cdot 0^2 - x^2}{x^2 - 2x \cdot 0 + 0^2} = \frac{-x^2}{x^2} = -1$$

along y-axis?

$$f(0,y) = \frac{0 \cdot y^2 - 0^2}{0^2 - 2 \cdot 0 y + y^2} = \frac{0}{y^2} = 0$$

$-1 \neq 0$, so the limit does not exist.

$f(x,y) = \frac{x^2 + \sin^2 y}{2y} e^{x+y}$ substitution gives $\frac{0}{0}$

along y-axis: $f(0,y) = \frac{0^2 + \sin^2 y}{2y} e^y = \frac{\sin^2 y \cdot e^y}{2y}$ where does this go as $y \to 0$? we get $\frac{0}{0}$ by substitution. try L'H:

$\lim_{y \to 0} \frac{\sin^2 y \, e^y}{2y} \overset{L'H}{=} \lim_{y \to 0} \frac{2\sin y \cdot \cos y \cdot e^y + \sin^2 y \, e^y}{2} =$

$= \lim_{y \to 0} \sin y (2\cos y \, e^y + \sin y \, e^y) = 0 \cdot (2 \cdot 1 \cdot 1 + 0 \cdot 1) = 0$

So $f(x,y) \to 0$ as $(x,y) \to (0,0)$ along y-axis.

along x-axis: $f(x,0) = \frac{x^2 + \sin^2 0}{2 \cdot 0} e^{x+0} = \frac{x^2 e^x}{0}$

this is undefined. we'll always be dividing by 0, regardless of x-value.

But, this is not enough for us to decide the limit dne!

we must find some other curve that gives a proper answer.

let's try $y = x^2$. $f(x,x^2) = \frac{x^2 + \sin^2(x^2)}{2x^2} e^{x+x^2}$

considering $\lim_{t \to 0} \frac{\sin t}{t} = 1$, write $\left(\frac{x^2}{2x^2} + \frac{\sin(x^2) \cdot \sin(x^2)}{2x^2} \right) e^{(x+x^2)}$

$= \left(\frac{1}{2} + \frac{1}{2} \cdot \frac{\sin(x^2)}{x^2} \sin(x^2) \right) e^{x+x^2}$

when we send $x \to 0$, this becomes $\left(\frac{1}{2} + \frac{1}{2} \cdot 1 \cdot 0 \right) e^0 = \frac{1}{2} \cdot 1 = \frac{1}{2}$

so, lim is $\frac{1}{2}$ along $y = x^2$ but. 0 along y-axis

Thus, the limit d.n.e.

$f(x,y) = \dfrac{x-y}{x+\sin y}$ substitution gives $\frac{0}{0}$

along x-axis $f(x,0) = \dfrac{x-0}{x+\sin(0)} = \dfrac{x}{x} = 1$

along y-axis $f(0,y) = \dfrac{0-y}{0+\sin y} = \dfrac{-y}{\sin y} \longrightarrow -1$ as $(x,y) \to (0,0)$

because $\lim_{t \to 0} \dfrac{\sin t}{t} = 1$ and $\lim_{t \to 0} \dfrac{t}{\sin t} = 1$ so $\lim_{t \to 0} \dfrac{-t}{\sin t} = -1$

$1 \neq -1$, so the limit does not exist.

$$f(x,y) = \frac{\arctan(x^2+y^2)}{1+e^{-x-y}}$$

$\arctan(0) = 0$ and $e^{-0-0} = e^0 = 1$

so substitution gives $\frac{0}{2}$. no problem!

$$\lim_{(x,y)\to(0,0)} \frac{\arctan(x^2+y^2)}{1+e^{-x-y}} = 0$$

$f(x,y) = x^2 y \ln(x^4 + y^2) - \pi$

Substitution causes errors because $\ln(t) \to -\infty$ as $t \to 0$

along x-axis $f(x,0) = x^2 \cdot 0 \cdot \ln(x^4 + 0^2) - \pi = 0 - \pi = -\pi$

along y-axis $f(0,y) = 0^2 \cdot y \cdot \ln(0^4 + y^2) - \pi = 0 - \pi = -\pi$

along $y = x^2$: $f(x, x^2) = x^2 x^2 \ln(x^4 + (x^2)^2) - \pi$

(I try this as it will combine the terms in argument of ln)

$= x^4 \ln(2x^4) - \pi$

let's try find $\lim_{x \to 0} x^4 \ln(2x^4) - \pi$

write $\lim_{x \to 0} \frac{\ln(2x^4) - x^{-4}\pi}{x^{-4}} \overset{L'H}{=} \lim_{x \to 0} \frac{\frac{1 \cdot 8x^3}{2x^4} + 4x^{-5}\pi}{-4x^{-5}}$

$= \lim_{x \to 0} \frac{\cancel{4} x^{-1} + \cancel{4} x^{-5}\pi}{-\cancel{4} x^{-5}} = \lim_{x \to 0} \frac{x^5(x^{-1} + x^{-5}\pi)}{-1} = \lim_{x \to 0} -x^4 - \pi = -\pi$

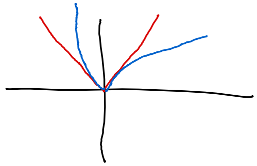
We suspect $\lim_{(x,y) \to (0,0)} x^2 y \ln(x^4 + y^2) - \pi$ is $-\pi = L$

(any other paths will also lead to doing L'Hopital)

Consider

$$0 \le \left| x^2 y \ln(x^4+y^2) - \pi - L \right| = \left| x^2 y \ln(x^4+y^2) - \pi + \pi \right|$$
$$= \left| x^2 y \ln(x^4+y^2) \right|$$

recall we said $\ln(1+t) \le t$, but is this helpful?
we can only say $|\ln(1+t)| \le |t|$ when t is positive.
plot $|\ln(1+t)|$ and $|t|$ to see:



if we try force that inequality we'll have trouble. we'd try by saying $\ln(x^4+y^2-1+1) \le x^4+y^2-1$ which is true, but we cannot put absolute values around it since $x^4+y^2-1 < 0$ when $(x,y) \to (0,0)$.
Thus, we'll need another approach.

$0 \le |x^2 y \ln(x^4+y^2)| = x^2|y|\,|\ln(x^4+y^2)|$

we know $\quad ab \le \frac{1}{2}(a^2+b^2)$ i.e. $2ab \le (a^2+b^2)$

so $\quad x^2|y| \le \frac{1}{2}(x^4+|y|^2) = \frac{1}{2}(x^4+y^2)$ (since $y^2=|y|^2$)

thus

$x^2|y|\,|\ln(x^4+y^2)| \le \frac{1}{2}(x^4+y^2)|\ln(x^4+y^2)|$

can I say anything about $\lim_{t\to 0^+} t|\ln t|$?

notice I only require $t\to 0^+$, since $x^4+y^2 \ge 0$

you may recall $\lim_{t\to 0^+} t\ln t = 0$, but I want to be extra sure for $\lim t|\ln t|$.

we know that $\lim_{t\to 0} |g(t)| = 0$ if and only if $\lim_{t\to 0} g(t) = 0$

we know $\lim_{t\to 0^+} t\ln t = 0$ so $\lim_{t\to 0^+} |t\ln t| = 0$

and since $t \ge 0$, $|t\ln(t)| = t|\ln t|$ and so

$\lim_{t\to 0^+} t|\ln(t)| = 0$.

Thus $\frac{1}{2}(x^4+y^2)|\ln(x^4+y^2)| \to \frac{1}{2}\cdot 0 = 0$

as $(x,y)\to(0,0)$.

Thus, by the squeeze theorem,

$\lim_{(x,y)\to(0,0)} x^2 y\ln(x^4+y^2) - \pi = L = -\pi$.

$f(x,y) = \sin(x)\ln(y)$

substitution causes issues as $\sin(0)=0$

but $\ln(y) \to -\infty$ as $y \to 0$.

along y-axis $f(0,y) = \sin(0)\ln(y) = 0 \cdot \ln(y)) = 0$

along x-axis $f(x,0) = \sin(x)\ln(0)$ ... $\ln(0)$ is not defined,

so f is not defined along y-axis

But this is not enough! if we could find some $y=g(x)$ s.t. $\ln(y) = \ln(g(x)) = \frac{1}{x}$, then we'd have $f(x,g(x)) = \frac{\sin x}{x}$ and we know that goes to 1 as $x \to 0$ (and $1 \neq 0$, which was the limit along the x-axis).

This is not hard! let $y = e^{1/x}$. then $\ln(y) = \frac{1}{x}$.

but how do we approach the origin (0,0) via this curve? notice if I send $x \to 0^-$, then $1/x \to -\infty$

so $e^{1/x} \to 0$, as $e^t \to 0$ as $t \to -\infty$.

thus we can approach $(0,0)$ along $y = e^{1/x}$ by sending $x \to 0^-$! Then $f(x, e^{1/x}) = \frac{\sin x}{x}$

and $\lim_{x \to 0^-} \frac{\sin x}{x} = 1$

Thus, we are sure the limit d.n.e.

In the next question, checking the limit exists is painful, simply because the expression is so complicated.

I include the checks for completeness, but we're told the limit is 0 in the notes, so you can skip ahead.

(does not mean I would not ask something like this in a test! I'd just tell you to skip the existence checks)

$$f(x,y) = \frac{x^3 y + 2\sin(xy^2)\cos(2x+y)}{x^2+y^2}$$

substitution leads to $0/0$.

along x-axis: $f(x,0) = \frac{x^3 \cdot 0 + 2\sin(x0^2)\cos(2x+0)}{x^2+0^2}$

$= \frac{0}{x^2} = 0$

along y-axis $f(0,y) = \frac{0^3 y + 2\sin(0 \cdot y^2)\cos(2 \cdot 0 + y)}{0^2+y^2} = \frac{0}{y^2} = 0$

along $x=y$: $f(x,x) = \frac{x^4 + 2\sin(x^3)\cos(3x)}{2x^2}$

leads to $0/0$ as $x \to 0$

Let's try L'H to find $\lim_{x\to 0} \frac{x^4 + 2\sin(x^3)\cos(3x)}{2x^2}$

$$\stackrel{L'H}{=} \lim_{x\to 0} \frac{4x^3 + 2\cos(x^3)\cdot 3x^2 \cdot \cos(3x) + 2\sin(x^3)\cdot(-\sin(3x))\cdot 3}{4x}$$

$$= \lim_{x\to 0} \frac{4x^3 + 6x^2\cos(x^3)\cos(3x) - 6\sin(x^3)\sin(3x)}{4x}$$

$$\stackrel{L'H}{=} \lim_{x\to 0} \left(\frac{-18x^4\cos(3x)\sin(x^3) - 18\cos(3x)\sin(x^3) - 36x^2\sin(3x)\cos(x^3) + 12x\cos(3x)\cos(x^3) + 12x^2}{4} \right)$$

(I used the internet to do that derivative.

finally, no division by x, so

$\lim_{x\to 0} = 0$.

we're convinced the limit exists and is 0.

now let's squeeze. This e.g. is excellent squeeze thm practice.

$$0 \le \left| \frac{x^3y + 2\sin(xy^2)\cos(2x+y)}{x^2+y^2} - 0 \right|$$

$$\le \left| \frac{x^3y}{x^2+y^2} \right| + \left| \frac{2\sin(xy^2)\cos(2x+y)}{x^2+y^2} \right|$$ since $|a+b| \le |a|+|b|$

$$= \frac{x^2|xy|}{x^2+y^2} + \left| \frac{2\sin(xy^2)\cos(2x+y)}{x^2+y^2} \right|$$

we know $\frac{x^2}{x^2+y^2} \le 1$

so $\frac{x^2|xy|}{x^2+y^2} \le |xy|$

and if

$|a| \le |b|$ then

$|a| + |c| \le |b| + |c|$

$$\le |xy| + \left| \frac{2\sin(xy^2)\cos(2x+y)}{x^2+y^2} \right|$$

$$= |xy| + \frac{2|\sin(xy^2)|\,|\cos(2x+y)|}{x^2+y^2}$$

$$\le |xy| + \frac{2|xy^2|\,|\cos(2x+y)|}{x^2+y^2}$$ since $|\sin(a)| \le |a|$, $|\sin(xy^2)| \le |xy^2|$

$$= |xy| + \frac{2|x|y^2|\cos(2x+y)|}{x^2+y^2}$$

$$\le |xy| + 2|x|\,|\cos(2x+y)|$$ $\frac{y^2}{x^2+y^2} \le 1$

$$\le |xy| + 2|x| \cdot 1$$ $|\cos(a)| \le 1$

$$= |x||y| + 2|x| \to 0 \text{ as } (x,y) \to (0,0)$$

Thus, by the squeeze theorem,

$$\lim_{(x,y)\to(0,0)} f(x,y) = 0.$$

$$f(x,y) = \frac{xy}{x^2 - xy + y^2}$$ substitution gives $\frac{0}{0}$

along x-axis $f(x,0) = \frac{x \cdot 0}{x^2 - x \cdot 0 + 0^2} = \frac{0}{x^2} = 0$

along y-axis $f(0,y) = \frac{0}{y^2} = 0$

along $y = x$ $f(x,x) = \frac{x^2}{x^2 - x^2 + x^2} = \frac{x^2}{x^2} = 1$

$1 \neq 0$ so limit dne.

$f(x,y) = \dfrac{\sin x}{\sqrt{x^2+y^2}}$ substitution gives $\frac{0}{0}$.

along x-axis: $f(x,0) = \dfrac{\sin x}{\sqrt{x^2}} = \dfrac{\sin x}{|x|}$

notice $\dfrac{\sin x}{|x|} \to 1$ if $x \geq 0$, since then we have $\lim\limits_{x \to 0^+} \dfrac{\sin x}{x}$. so along positive x-axis, $f \to 1$.

but if $x \leq 0$, then $\sin x \leq 0$ too, but $|x|$ still pos!

so $\dfrac{\sin x}{|x|} \to -1$ if we approach along the neg x-axis.

Since $-1 \neq 1$, limit does not exist

(also along y-axis from either direction, it's 0)

✓

$f(x,y) = (\cos(x) - 1)\ln(y)$

substitution causes issues as $\ln(y) \to -\infty$ as $y \to 0$

along y-axis: $f(0,y) = (\cos 0) - 1)\ln y$
$= (1-1)\ln y = 0 \ln y = 0$

along x-axis $f(x,0) = (\cos x - 1)\ln(0)$

undefined along x-axis as $\ln(0)$ is undefined

But this is not enough. Need to find an approach to the origin that gives a defined answer.

we'll take a similar approach to when we faced $\sin x \ln y$ — we want to get rid of the $\ln(y)$ and form $\frac{\cos x - 1}{\text{something}}$ s.t. we can handle it.

of course, we need to be able to approach the origin along the curve we choose. (the curve I'm about to choose would never come up in a test, please don't stress. This is an illustrative exercise!)

let $y = e^{-1/x^2}$. firstly, can I approach the origin?

indeed. if $x \to 0$ from either side, $\frac{1}{x^2} \to \infty$ so $-1/x^2 \to -\infty$ so $e^{-1/x^2} \to 0$ since $e^t \to 0$ as $t \to -\infty$.

great!

Thus, $f(x, e^{-1/x^2}) = (\cos x - 1)\ln(e^{-1/x^2}) = (\cos x - 1)\frac{-1}{x^2}$

$= \frac{1 - \cos x}{x^2}$. where does this go as $x \to 0$?

$$\lim_{x \to 0} \frac{1 - \cos x}{x^2} \overset{L'H}{=} \lim_{x \to 0} \frac{\sin x}{2x} = \frac{1}{2} \lim_{x \to 0} \frac{\sin x}{x} = \frac{1}{2} \cdot 1 = \frac{1}{2}$$

recall $f \to 0$ along y-axis. and $0 \neq \frac{1}{2}$.

Thus, limit dne!

$$f(x,y) = \frac{x^{1/4} y^{7/3}}{x^2+y^2}$$ substitution: $\frac{0}{0}$.

along either axis we'll get $\frac{0}{x^2}$ or $\frac{0}{y^2}$ so 0.

along $x=y$ we get $f(x,x) = \frac{x^{(1/4+7/3)}}{2x^2}$

$$= \frac{x^{(3/12+\frac{28}{12})}}{2x^2} = \frac{x^{31/12}}{2x^2} = \frac{x^{(\frac{24}{12}+\frac{7}{12})}}{2x^2} = \frac{x^2 x^{7/12}}{2x^2} =$$

$$\frac{x^{7/12}}{2} \to 0 \text{ as } (x,y) \to (0,0)$$

let's try squeeze theorem w/ $L=0$

$$0 \le \left| \frac{x^{1/4} y^{7/3}}{x^2+y^2} - 0 \right| = \frac{|x^{1/4} y^{\left(\frac{6}{3}+\frac{1}{3}\right)}|}{x^2+y^2}$$

$$= \frac{|x^{1/4} \cdot y^2 \cdot y^{1/3}|}{x^2+y^2} = \frac{y^2 |x^{1/4}||y^{1/3}|}{x^2+y^2} \le |x^{1/4}||y^{1/3}|$$

since $\frac{y^2}{x^2+y^2} \le 1$.

$|x^{1/4}||y^{1/3}| \to 0$ as $(x,y) \to (0,0)$ so

by the squeeze thm,

$$\lim_{(x,y)\to(0,0)} \frac{x^{1/4} y^{7/3}}{x^2+y^2} = 0$$

$f(x,y) = \frac{\cos(2x)-1}{\sqrt{x^2+y^2}}$ substitution: $\frac{1-1}{0} = \frac{0}{0}$

along x-axis: $f(x,0) = \frac{\cos(2x)-1}{\sqrt{x^2}}$ again $\frac{0}{0}$ ... let's use L'H: to find out what happens when $x \to 0$

$\lim_{x \to 0} \frac{\cos(2x)-1}{\sqrt{x^2}} = \lim \frac{\cos(2x)-1}{(x^2)^{1/2}}$

$= \lim_{x \to 0} \frac{-2\sin(2x)}{\frac{1}{2}(x^2)^{-1/2}} = \lim_{x \to 0} \frac{-2\sin(2x)\cdot(x^2)^{1/2}}{1/2}$

$= \lim_{x \to 0} -4\sin(2x)\sqrt{x^2} = 0$

so $f \to 0$ along x-axis as $(x,y) \to (0,0)$

along y-axis: $f(0,y) = \frac{\cos(0)-1}{\sqrt{0+y^2}} = \frac{1-1}{\sqrt{y^2}} = \frac{0}{\sqrt{y^2}} = 0$

along $y=x$: $f(x,x) = \frac{\cos(2x)-1}{\sqrt{2x^2}} = \frac{\cos(2x)-1}{\sqrt{2}\sqrt{x^2}}$ $0/0$ let's L'H again

$\lim_{x \to 0} \frac{\cos(2x)-1}{\sqrt{2}\sqrt{x^2}} \overset{L'H}{=} \lim_{x \to 0} \frac{-2\sin(2x)}{\sqrt{2}\frac{1}{2}(x^2)^{-1/2}} = \lim \frac{-2\sin(2x)\sqrt{x^2}}{\sqrt{2}/2}$

$= 0$ again.

so we suspect $L = 0$

Now, this question gave me so much trouble!! I'm leaving in my rough work to illustrate that these questions may take multiple attempts

$$0 \le \left| \frac{\cos(2x)-1}{\sqrt{x^2+y^2}} \right| = \frac{|\cos(2x)-1|}{\sqrt{x^2+y^2}}$$

splitting this up does not help:

$$\le \frac{|\cos 2x|}{\sqrt{x^2+y^2}} + \frac{1}{\sqrt{x^2+y^2}} \quad \text{since } |a-b| \le |a|+|b|$$

$$\le \frac{1}{\sqrt{x^2+y^2}} + \frac{1}{\sqrt{x^2+y^2}} \quad \text{since } |\cos(a)| \le 1$$

$$= \frac{2}{\sqrt{x^2+y^2}} \quad \text{not helpful.}$$

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trig identity perhaps?

$$\cos(2x) = 2\cos x \sin x$$

$$\frac{|\cos 2x - 1|}{\sqrt{x^2+y^2}} = \frac{|2\cos x \sin x - 1|}{\sqrt{x^2+y^2}}$$

$1 = \cos^2 x + \sin^2 x$

so $|2\cos x \sin x - 1|$

$= |1 - \cos^2 x - \sin^2 x + 2\cos x \sin x|$

$= |\cos^2 x + \sin^2 x - 2\cos x \sin x|$

$= |(\cos x - \sin x)^2|$

$$= \frac{(\cos x - \sin x)^2}{\sqrt{x^2+y^2}} \quad \text{now what?}$$

---

maybe change of variables?

$x = r\cos\theta, \quad y = r\sin\theta$

$$= \frac{|\cos(2r\cos\theta) - 1|}{r}$$

still, going to get $\frac{1+1}{r}$

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can I convert $\cos(a)$ to sin? yes

$$\cos(a) = \sin\left(\frac{\pi}{2} - a\right)$$

so $\cos(2x) = \sin\left(\frac{\pi}{2} - 2x\right)$

then $\left|\sin\left(\frac{\pi}{2} - 2x\right)\right| \le \left|\frac{\pi}{2} - 2x\right|$

$$\frac{|\cos(2x)-1|}{\sqrt{x^2+y^2}} = \frac{|\sin(\frac{\pi}{2}-2x) - 1|}{\sqrt{x^2+y^2}} \le \frac{|\sin(\frac{\pi}{2}-2x)| + 1}{\sqrt{x^2+y^2}}$$

$$\le \frac{|\frac{\pi}{2} - 2x|}{\sqrt{x^2+y^2}} + \frac{1}{\sqrt{x^2+y^2}} \le \frac{\frac{\pi}{2}}{\sqrt{x^2+y^2}} + \frac{2|x|}{\sqrt{x^2+y^2}} + \frac{1}{\sqrt{x^2+y^2}}$$

$\frac{|x|}{\sqrt{x^2+y^2}} \le 1$ so

$$\le \frac{\frac{\pi}{2}}{\sqrt{x^2+y^2}} + 2 + \frac{1}{\sqrt{x^2+y^2}} = 2 + \frac{\frac{\pi}{2}+1}{\sqrt{x^2+y^2}}$$

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from checking desmos I see

$|\cos(2x) - 1| \le |x|$ for small x

so $\frac{|\cos(2x)-1|}{\sqrt{x^2+y^2}} \le \frac{|x|}{\sqrt{x^2+y^2}} \le 1$ not helpful.

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get rid of square root?

$$\frac{|\cos(2x)-1|}{\sqrt{x^2+y^2}} = \frac{\sqrt{x^2+y^2}\,|\cos(2x)-1|}{x^2+y^2}$$

???

this is frustrating to the point where I had to check the graph on geogebra to make sure the limit is indeed 0. It is.

can I get rid of y at least to have
$0 \leq |f(x,y) - 0| \leq g(x)$, and then perhaps
$\lim_{x \to 0} g(x) = 0$? $\lim g(x)$ may require work to show

$x^2 \leq x^2 + y^2$ so $\sqrt{x^2} \leq \sqrt{x^2+y^2}$ so $\frac{1}{\sqrt{x^2+y^2}} \leq \frac{1}{\sqrt{x^2}}$

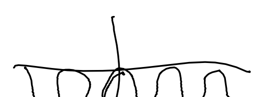
thus $\frac{|\cos(2x)-1|}{\sqrt{x^2+y^2}} \leq \frac{|\cos(2x)-1|}{\sqrt{x^2}}$ ha!

If I can show

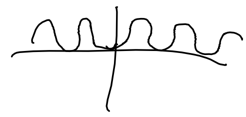
$\lim_{x \to 0} \frac{|\cos(2x)-1|}{\sqrt{x^2}} = 0$, then I'm done!

let's think about $|\cos(2x)-1|$:

$\cos(2x)-1$:



$|\cos(2x)-1|$:



which is just

$1-\cos(2x)$ (no absolute vals!)

$\lim_{x \to 0} \frac{|\cos(2x)-1|}{\sqrt{x^2}} = \lim_{x \to 0} \frac{1-\cos(2x)}{\sqrt{x^2}} \stackrel{L'H}{=}$

$= \lim_{x \to 0} \frac{2\sin(2x)}{\frac{1}{2}(x^2)^{-1/2}} = \lim_{x \to 0} 4\sin(2x)\sqrt{x^2} = 0$!!!

finally, by the squeeze theorem,

$\lim_{(x,y) \to (0,0)} \frac{\cos(2x)-1}{\sqrt{x^2+y^2}} = 0$

$$f(x,y) = \frac{\sin(x^3+y^3)}{x^2+y^2}$$ substitution: $\frac{0}{0}$

along x-axis: $f(x,0) = \frac{\sin(x^3)}{x^2}$ leads to $\frac{0}{0}$

$$\lim_{x\to 0} \frac{\sin(x^3)}{x^2} \overset{L'H}{=} \lim_{x\to 0} \frac{3x^{\cancel{2}}\cos(x^3)}{2\cancel{x}} = \lim_{x\to 0} \frac{3}{2}\cdot x \cdot \cos(x^3)$$

$$= 0\cdot 1 = 0$$

along y-axis: $f(0,y) = 0$ by symmetry (just replace x with y above)

along $y=x$: $f(x,x) = \frac{\sin(2x^3)}{2x^2}$ again $\frac{0}{0}$

$$\lim_{x\to 0} \frac{\sin(2x^3)}{2x^2} \overset{L'H}{=} \lim \frac{6x^{\cancel{2}}\cos(2x^3)}{4\cancel{x}} = \lim_{x\to 0} \frac{6x}{4}\cos(2x^3)$$

$$= 0\cdot 1 = 0$$

so we suspect lim $= 0$

let's squeeze

$$0 \le \left| \frac{\sin(x^3+y^3)}{x^2+y^2} - 0 \right| = \frac{|\sin(x^3+y^3)|}{x^2+y^2}$$

$$\le \frac{|x^3+y^3|}{x^2+y^2} \quad \text{since} \quad |\sin(a)| \le |a|$$

$$\le \frac{|x^3|}{x^2+y^2} + \frac{|y^3|}{x^2+y^2} \quad \text{since } |a+b| \le |a|+|b|$$

$$= \frac{x^2|x|}{x^2+y^2} + \frac{y^2|y|}{x^2+y^2}$$

$\frac{x^2}{x^2+y^2} \le 1$

so $\frac{x^2|x|}{x^2+y^2} \le |x|$

$$\le |x| + \frac{y^2|y|}{x^2+y^2}$$

so $\frac{x^2|x|}{x^2+y^2} + \frac{y^2|y|}{x^2+y^2} \le |x| + \frac{y^2|y|}{x^2+y^2}$

$$\le |x| + |y|$$

$$\to 0$$

again, $\frac{y^2}{x^2+y^2} \le 1$ so

$\frac{y^2|y|}{x^2+y^2} \le |y|$ so

$|x| + \frac{y^2|y|}{x^2+y^2} \le |x| + |y|$

thus, by the squeeze thm

$$\lim_{(x,y)\to(0,0)} \frac{\sin(x^3+y^3)}{x^2+y^2} = 0$$

$$f(x,y) = \frac{|\sin y|\, x}{x^2 + \sqrt{|y|}}$$

along x-axis $f(x,0) = \dfrac{|\sin 0|\, x}{x^2 + 0} = \dfrac{0}{x^2} = 0$

along y-axis $f(0,y) = \dfrac{|\sin y|\, 0}{\sqrt{|y|}} = \dfrac{0}{\sqrt{|y|}} = 0$

I can't see any other curves causing problems, and we're anyway told $\lim = 0$

let's squeeze...

$$0 \le \left| \frac{|\sin y|\, x}{x^2+\sqrt{|y|}} - 0 \right| = \frac{|\sin y|\,|x|}{x^2+\sqrt{|y|}}$$ since $|\sin y|$, x^2, $\sqrt{|y|}$ are all already ≥ 0.

$$= \frac{|\sin y|\,|x|}{x^2+\sqrt{|y|}} \le \frac{|y|\,|x|}{x^2+\sqrt{|y|}}$$ since $|\sin a| \le |a|$.

$ab \le \frac{1}{2}(a^2+b^2)$

so $|a||b| \le \frac{1}{2}(|a|^2+|b|^2)$
$= \frac{1}{2}(a^2+b^2)$

$$\le \frac{\frac{1}{2}(y^2+x^2)}{x^2+\sqrt{|y|}}$$

$$= \frac{\frac{1}{2}y^2}{x^2+\sqrt{|y|}} + \frac{\frac{1}{2}x^2}{x^2+\sqrt{|y|}}$$

$x^2 \le x^2 + \sqrt{|y|}$ since $\sqrt{|y|} \geq 0$

so $\frac{x^2}{x^2+\sqrt{|y}} \le 1$

$$\le \frac{\frac{1}{2}y^2}{x^2+\sqrt{|y|}} + \frac{1}{2}$$

$$= \frac{\frac{1}{2}y^{3/2}\cdot\sqrt{y}}{x^2+\sqrt{|y|}} + \frac{1}{2}$$

$y \le |y|$ (either equal, or y is neg)

so $\sqrt{y} \le \sqrt{|y|}$

so $\sqrt{y} \le \sqrt{|y|} + x^2$

so $\frac{\sqrt{y}}{\sqrt{|y|}+x^2} \le 1$

$$\le \frac{1}{2}y^{3/2} + \frac{1}{2}$$

aaand, that goes to $\frac{1}{2}$, so not helpful!
let's try again $\rightarrow$

$$0 \le \left| \frac{|\sin y| x}{x^2 + \sqrt{|y|}} \right|$$

turning this into 2 fractions did not help, let's keep it as 1.

$$= \frac{|\sin y|\,|x|}{x^2 + \sqrt{|y|}} \le \frac{|y|\,|x|}{x^2 + \sqrt{|y|}}$$ since $|\sin y| \le |y|$!

what can we say about $|y|$ and $\sqrt{|y|}$?

for a positive, if $0 \le a \le 1$, $a \le \sqrt{a}$, right? (e.g. $\frac{1}{4} < \sqrt{\frac{1}{4}} = \frac{1}{2}$)

so for small enough y, which is the case when $y \to 0$, we could say $|y| \le \sqrt{|y|}$

then $|y| \le \sqrt{|y|} + x^2$ since $x^2 \ge 0$, so

$$\frac{|y|}{\sqrt{|y|} + x^2} \le 1$$ useful!

thus $$\frac{|x|\,|y|}{\sqrt{|y|} + x^2} \le |x| \to 0$$

So by the squeeze theorem,

$$\lim_{(x,y)\to(0,0)} \frac{|\sin y| x}{x^2 + \sqrt{|y|}} = 0.$$